

Hyperscale Green Data Center Sahara Neural Hub

Sovereign African Data Storage & Decentralized High-Performance
Computing — Dakhla, Morocco

Comprehensive feasibility analysis for a 50 MW AI-ready hyperscale data center powered by 100% renewable energy (wind + solar + BESS), leveraging Dakhla's world-class wind resources (LCOE \$0.022/kWh), operational submarine cable connectivity (20-40 Tbps to Europe & West Africa), and Dakhla West Africa Free Zone tax advantages. Total estimated CAPEX: \$611M - \$1.135B. Projected IRR: 12-18%.

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1. Executive Summary

This report presents a comprehensive feasibility analysis for the construction and operation of a **Hyperscale Green Data Center** in Dakhla, Morocco, dedicated to **Sovereign African Data Storage** and **Decentralized High-Performance Computing (HPC)** powered by artificial intelligence GPU clusters. The project, codenamed "**Sahara Neural Hub**", aims to establish a 50 MW IT-load facility powered entirely by renewable energy (wind and solar), leveraging the exceptional wind resources of the Dakhla-Oued Ed-Dahab region, which ranks among the top five onshore wind sites globally with average wind speeds of 9.5-10.8 m/s at 80m height.

The strategic positioning of Dakhla offers a unique convergence of advantages: abundant and inexpensive renewable energy (LCOE as low as \$0.022-0.035/kWh), an operational submarine cable landing station providing 20-40 Tbps connectivity to both Europe and West Africa, a favorable tax regime within the Dakhla West Africa Free Zone (5-year corporate tax exemption followed by a reduced 15% rate, VAT and customs duty exemptions), and a cool coastal desert climate enabling 5,000-7,500 hours per year of free cooling. The total estimated CAPEX ranges from \$611 million to \$1.135 billion for a 50 MW facility including 2,000 NVIDIA H100 GPUs, with annual OPEX projected at \$82-99 million.

The African data center market is experiencing explosive growth, projected to reach USD 6.81 billion by 2030 at a CAGR of 11.8%. Morocco alone commands 35% of all new announced power capacity on the continent, with a pipeline exceeding 1,886 MW. No data center currently exists in Dakhla, presenting a compelling first-mover advantage for a facility that can serve as the strategic gateway between European and West African digital infrastructure, addressing the growing demand for sovereign African data storage driven by data localization laws in 36+ African countries and the African Union Data Policy Framework adopted in February 2024.

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Table 1: Project Key Metrics Summary

2. Market Analysis

2.1 African Data Center Market Overview

The African data center market represents one of the most dynamic growth opportunities in global digital infrastructure. Valued at USD 3.49 billion in 2024, the market is projected to reach USD 6.81 billion by 2030, growing at a compound annual growth rate (CAGR) of 11.8%. The construction segment alone is expected to expand from USD 1.26 billion in 2025 to USD 3.06 billion by 2030, reflecting the massive infrastructure buildout underway across the continent. The Middle East and Africa colocation market is experiencing even more rapid growth, reaching USD 4.9 billion in 2026 with a 28.5% year-over-year increase, and is projected to hit USD 11.1 billion by 2030. Total existing data center capacity across the top five African markets exceeds 500 MW, with 56 new data centers and approximately 400 MW of additional capacity announced by the end of 2025.

Country	Data Centers	Capacity (MW)	Key Notes
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Table 2: African Data Center Market by Country (2025)

2.2 Data Sovereignty and Regulatory Drivers

The push for data sovereignty in Africa has become a major driver of data center demand. As of 2025, more than 36 African countries have enacted or are actively developing data protection legislation, up from fewer than 10 in 2015. The African Union Data Policy Framework, adopted in February 2024, explicitly supports the development of regional and continental data infrastructure while advising against overly broad data localization mandates. Kenya requires that a copy of personal data be stored domestically, Nigeria mandates localization for certain sectors including financial data, and Egypt has introduced localization requirements for specific data categories. Morocco's Loi 09-08 (2009) governs data protection with cross-border transfers allowed subject to equivalent protection standards, and the country is actively modernizing its framework to align more closely with the EU's GDPR. This growing regulatory landscape creates substantial demand for sovereign African data storage facilities that can serve clients who need to keep data on the continent while still accessing world-class infrastructure and connectivity.

2.3 AI and GPU Computing Demand in Africa

The artificial intelligence sector in Africa is experiencing rapid expansion, with over 2,360 companies listing AI as a specialization as of 2022, and the continent's cloud computing adoption growing at 25-30% annually. However, the supply of AI-ready data center infrastructure severely lags demand: in South Africa, only 5 out of 56 operational data centers are equipped for AI workloads. AI-ready data centers consume approximately 5 times more electricity than traditional cloud facilities, and GPU-based racks require 40-140 kW per rack compared to 5-12 kW for conventional IT racks. This 10-30x increase in power density represents both a challenge and an opportunity for a purpose-built green HPC facility. The African Union Continental AI Strategy, adopted in July 2024, further reinforces the imperative for local AI compute capacity. Currently, only about 17% of African enterprises have fully migrated to cloud, indicating massive untapped potential as digital transformation

accelerates across the continent.

2.4 Morocco's Competitive Position

Morocco has emerged as the leading destination for data center investment in Africa, commanding 35% of all newly announced power capacity on the continent. The kingdom's pipeline includes the Iozera project (386 MW in Tetouan, \$500M investment, MoU signed 2024), the Naver Cloud facility (500 MW in Tetouan, partnership with NVIDIA, construction starting Q4 2025), and the Nouaceur project (500 MW target near Casablanca, 100% renewable energy planned). Morocco's advantages include proximity to Europe (only 15 km at the Strait of Gibraltar), submarine cable landings with over 500 Tbps total design capacity, a renewable energy share of 44% in the energy mix (highest in North Africa), a stable constitutional monarchy with consistent pro-business policies, and a bilingual workforce fluent in French, Arabic, and increasingly English. The "Digital Morocco 2030" strategy targets 40 billion MAD in revenue from advanced digital services by 2030, signaling strong government commitment to the digital infrastructure sector.

3. Location Analysis: Dakhla

3.1 Geographic and Climatic Advantages

Dakhla is situated at coordinates 23 degrees 41 minutes North, 15 degrees 57 minutes West on a narrow peninsula extending into the Atlantic Ocean, approximately 1,500 km south of Casablanca. The city benefits from a unique microclimate shaped by the cold Canary Current, with average temperatures of 20-22 degrees Celsius year-round, consistently low humidity, and steady ocean breezes. These conditions are exceptionally favorable for data center operations, enabling 5,000-7,500 hours per year of free cooling through air-side economization, which can reduce cooling energy consumption by 60-80% compared to traditional mechanical refrigeration. The Atlantic coast location also opens the possibility of seawater cooling, with the potential to achieve a Power Usage Effectiveness (PUE) as low as 1.05-1.15, placing the facility among the most energy-efficient data centers globally. For context, Google's best-in-class facilities achieve a PUE of 1.10, and the industry average stands at 1.40-1.60.

3.2 Renewable Energy Resources

The Dakhla-Oued Ed-Dahab region possesses the highest wind energy density in Morocco and ranks among the top five onshore wind sites globally. At 80m height, the average wind speed ranges from 9.5 to 10.8 m/s with a power density of approximately 480 W/m², driven by consistent northeasterly trade winds associated with the Azores anticyclone. The LCOE for wind energy in Dakhla is estimated at \$0.022-0.035/kWh, significantly below the global weighted average of \$0.034/kWh (IRENA 2024) and well below Morocco's average electricity tariff of \$0.116/kWh for business users. Morocco's national wind capacity reached 2,373 MW by the end of 2024, and the country targets 52% renewable installed capacity by 2030. A 60 MW wind farm in Dakhla began injecting power in March 2025, with a 40 MW expansion coupled to the desalination plant. A critical infrastructure project, the Dakhla-Casablanca 400 kV high-voltage transmission line with 3 GW capacity, is under development with delivery expected by 2028, which will be essential for evacuating surplus renewable energy production.

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Table 3: Dakhla Renewable Energy Resources

3.3 Connectivity and Submarine Cables

Dakhla possesses a strategic advantage that few African cities can match: it is already a submarine cable landing point. The Maroc Telecom West Africa cable, operational since 2021, lands directly in Dakhla with a capacity of 20 Tbps (extensible to 40 Tbps), connecting Morocco to Mauritania, Senegal, Cote d'Ivoire, Togo, Benin, and Gabon. This positions Dakhla as a natural gateway between European and West African digital traffic. Additionally, the Sahara cable (60 Tbps) is being deployed by Maroc Telecom, the Islalink Canaries-Morocco cable (EUR 49M, delivery 2028) will connect southern Morocco to the Canary Islands and onward to Europe, and the MEDUSA cable (480 Tbps total, landed in Nador December 2025) provides additional Mediterranean connectivity. Morocco's total accessible international bandwidth exceeds 500 Tbps in design capacity. Latency from Casablanca to Paris is approximately 25-30 ms, and Dakhla's direct West Africa cable link provides low-latency access to key West African markets. Fiber-to-the-home (FTTH) is available in Dakhla via Maroc Telecom, Orange, and Inwi at speeds up to 1 Gbps.

Cable System	Capacity	Status	Connects To
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Table 4: Submarine Cable Connectivity from Dakhla/Morocco

3.4 Land and Real Estate

Land in Dakhla is remarkably affordable compared to Morocco's northern cities, offering substantial cost savings for a hyperscale facility requiring large land areas. Constructible land in the PK40 development zone starts at 400 MAD/m² (approximately 37 EUR/m²), which represents a 30-40% discount compared to Casablanca and Marrakech. Industrial zone lots are available at even lower prices (14-16 MAD/m² for small lots), and the Dakhla West Africa Free Zone encompasses 1,000 hectares of industrial land. A new 27-hectare industrial zone with 326 lots has been announced with a 30 MAD million development investment. For a 50 MW data center, approximately 10 hectares would be required for the facility buildings, electrical infrastructure, and wind/solar energy installations. At PK40 prices, this represents a land acquisition cost of approximately 40 MAD million (\$4 million), which is negligible relative to the total project CAPEX. Rural land in the broader Dakhla-Oued Ed-Dahab region can be obtained for as little as 5-50 MAD/m² through the Agence de Developpement Agricole, which has 1,778 hectares available for long-term lease. Additionally, over 11,120 hectares were mobilized from the private domain of the state in 2023, the largest allocation of any region in Morocco, indicating strong government commitment to facilitating investment.

Zone Type	Price (MAD/m ²)	Price (EUR/m ²)	Notes
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Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'PK40 Lotissements' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='PK40 Lotissements', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '400 - 625' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='400 - 625', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '37 - 58' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='37 - 58', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Developing zone' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Developing zone', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
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Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'DWAFZ (Free Zone)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='DWAFZ (Free Zone)', textColor=Color (.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Administrative allocation' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Libera tionSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Administrative allocation', textColor=Col or(.117647,.121569,.1294 12,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': " 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '1,000 hectares' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='1,000 hectares', textColor =Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

Table 5: Land Prices in Dakhla by Zone Type

3.5 Water and Desalination Infrastructure

Water availability is a critical consideration for data center operations, particularly for cooling systems. Dakhla, situated in an arid region, addresses this challenge through a major desalination infrastructure program. A new seawater desalination station with a capacity of 37 million m³/year (approximately 101,000 m³/day) is under construction with a total investment of 2.5 billion MAD (approximately EUR 230 million), of which 7 million m³/year is allocated to drinking water and 30 million m³/year to irrigation for 5,200 hectares. The plant is 100% powered by a coupled wind farm, reducing the cost per cubic meter by approximately 30% compared to standard desalination, yielding an estimated cost of 5-7 MAD/m³. An existing brackish water desalination station with a capacity of 17,000 m³/day has been operational since 2015. For a 50 MW data center using advanced cooling technologies (direct liquid cooling and free cooling), water consumption would be minimal compared to traditional evaporative cooling systems, estimated at 500-1,000 m³/day for make-up water, well within the capacity of the new desalination infrastructure. The possibility of seawater cooling using deep ocean water (4-6 degrees Celsius at 700-1,000m depth) could eliminate the need for evaporative cooling entirely, achieving a PUE of 1.05-1.15.

3.6 Electrical Grid and Power Infrastructure

Dakhla's electrical infrastructure has undergone significant upgrades in recent years. The city is connected to Morocco's national grid via a 400 kV double-circuit transmission line, with a total investment of approximately 2.4 billion MAD. Current installed generation capacity includes a 76 MW diesel power station, a 60 MW wind farm (first injections in March 2025), and a 40 MW wind installation coupled to the desalination project. The most critical infrastructure project is the Dakhla-Casablanca 1,000 km ultra-high-voltage transmission corridor with a planned capacity of 3,000 MW (1,500 MW in Phase 1, expansion to 3,000 MW by 2028), which will enable the evacuation of 900 MW of additional renewable energy production from the southern provinces. Five candidates have been shortlisted for the construction contract as of February 2025. For a 50 MW data center, the power requirement with a PUE of 1.2 would be approximately 60 MW of total electrical input. This would require a dedicated renewable energy installation of approximately 120-150 MW (wind + solar + BESS) to ensure 24/7 power availability, given the intermittent nature of renewable sources. The cost of this energy infrastructure is estimated at \$50-100 million for the renewable generation component, with a payback period of 7-12 years based on current electricity tariff differentials.

4. Technical Architecture

4.1 Data Center Design Specifications

The Sahara Neural Hub is designed as a Tier III+ hyperscale facility optimized for GPU-accelerated AI workloads. The design philosophy centers on three pillars: energy efficiency through green design (target PUE 1.10-1.20), high-density computing (40-140 kW per rack for GPU clusters), and sovereign data assurance (compliance with Moroccan Loi 09-08, AU Data Policy Framework, and GDPR-equivalent standards). The facility will be constructed in two phases: Phase 1 (24-36 months) delivering 25 MW IT load with 1,000 GPU servers, and Phase 2 (subsequent 18-24 months) expanding to 50 MW with 2,000+ GPUs. The data hall design supports both air-cooled racks (up to 15 kW for traditional IT) and liquid-cooled racks (up to 140 kW for NVIDIA GB200 NVL72 systems), with direct-to-chip liquid cooling as the primary cooling method for GPU clusters and rear-door heat exchangers as a secondary system. The facility will use 100% renewable energy sourced from a dedicated wind farm and solar PV installation, backed by battery energy storage systems (BESS) and grid connection for reliability. Seawater cooling is under evaluation as a tertiary cooling method, leveraging Dakhla's Atlantic coast location.

Specification	Phase 1	Phase 2 (Full Build)
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Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Data Hall Area' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Data Hall Area', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '5,000 m²' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='5,000 m', textColor=Color(.117647,.121569,.129412,1), us_lines=[]), ParaFrag(__tag__='super', bold=0, fontName='LiberationSerif', fontSize=7.2, greek=0, italic=1, link=[], rise=4.5, text='2', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '10,000 m²' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='10,000 m', textColor=Color(.117647,.121569,.129412,1), us_lines=[]), ParaFrag(__tag__='super', bold=0, fontName='LiberationSerif', fontSize=7.2, greek=0, italic=1, link=[], rise=4.5, text='2', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
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Table 6: Data Center Design Specifications

4.2 GPU Infrastructure and AI Compute

The computational backbone of the Sahara Neural Hub consists of NVIDIA H100 and next-generation B200 GPU clusters interconnected via NVIDIA's Quantum-2 InfiniBand network. The H100 SXM5 GPU is priced at \$25,000-40,000 per unit with a maximum TDP of 700W, while the DGX H100 system (8x H100) costs \$300,000-450,000 and consumes 10.2 kW. The newer B200 SXM6 GPU is priced at \$30,000-40,000 with the DGX B200 at approximately \$515,000 consuming 14.3 kW. The GB200 NVL72 rack system (72 Blackwell GPUs, 13.5 TB GPU RAM) costs approximately \$3 million per rack and draws 120-140 kW, requiring mandatory liquid cooling. The facility's InfiniBand fabric will use NVIDIA Quantum-2 QM9700 switches (64-port NDR 400Gb/s, approximately \$35,000 list price) in a fat-tree topology to ensure non-blocking inter-rack communication at 400 Gb/s per GPU. Cloud GPU pricing benchmarks show AWS H100 instances at \$6.88/GPU-hour (on-demand) and specialized neo-cloud providers at \$2.01-3.44/GPU-hour, providing a revenue floor for the facility's GPU-as-a-Service offering. At these rates, a fully utilized 2,000-GPU cluster could generate \$35-120 million in annual GPU compute revenue.

Component	Unit Price (USD)	Power (W)	Notes
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Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'DGX B200 (8x GPU)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='DGX B200 (8x GPU)', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '~\$515,000' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text=~\$515,000', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '14,300' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='14,300', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '8-GPU Blackwell node' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='8-GPU Blackwell node', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

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Table 7: GPU Infrastructure Component Pricing

4.3 Green Energy System Architecture

The energy system is designed as a hybrid renewable microgrid combining wind turbines, solar photovoltaic arrays, and battery energy storage systems (BESS) with grid backup. For a 50 MW IT load with a PUE of 1.2 (60 MW total input), the renewable installation would comprise approximately 80-100 MW of wind capacity (capitalizing on Dakhla's exceptional 45% capacity factor) and 40-60 MW of solar PV, complemented by a 200 MWh BESS system providing 4 hours of storage at critical load. The CAPEX for this renewable energy installation is estimated at \$100-200 million based on current costs: wind at \$1,100-1,500/kW, solar PV at \$600-900/kW, and BESS at \$117-300/kWh (turnkey utility-scale 4-hour systems). The wind component alone, at 100 MW, would consist of approximately 20-25 turbines of the Vestas V136-4.5 MW or Siemens Gamesa SG 4.5-145 class, with delivery timelines of 12-18 months from order. The OPEX for the wind farm is estimated at \$42,000-48,000/MW/year. The system would be designed for self-consumption with grid export of surplus energy, leveraging Morocco's Law 58-15 which authorizes the sale of renewable energy surplus to the ONEE grid under a net billing mechanism. The ANRE (National Electricity Regulatory Authority) is finalizing the tariff framework with a reform expected by 2027.

4.4 Cooling Architecture

The cooling strategy employs a three-tier approach optimized for Dakhla's climate. The primary system is direct-to-chip liquid cooling (DLC) for GPU racks, which removes 60-80% of heat directly from the chip surface using cold plates and a closed-loop water circuit, with a cost premium of 15-25% over air cooling infrastructure. The secondary system leverages Dakhla's ambient conditions for air-side free cooling during the 5,000-7,500 hours per year when outdoor temperatures fall below the supply air setpoint, using economizer dampers to bring filtered outside air directly into the data hall. The tertiary system under evaluation is seawater cooling, utilizing the cold Atlantic waters at a depth of 700-1,000 meters (4-6 degrees Celsius) through a heat exchanger system, which could reduce cooling energy consumption by 80-90% and achieve a PUE of 1.05-1.15. The CAPEX for seawater cooling is estimated at \$80-200/kW (thermal), with potential cost reduction to \$10-20/kW at 1,000 MWt scale. For air-cooled racks (up to 15 kW), rear-door heat exchangers at \$3,000-8,000 per rack provide an efficient retrofit option. The overall cooling system design targets a PUE contribution of 0.10-0.15 from the cooling infrastructure, compared to 0.40-0.60 for traditional data centers, representing an annual energy saving of approximately 26 million kWh for a 10 MW reduction in PUE overhead, valued at approximately \$2.6 million per year at \$0.10/kWh.

5. Financial Analysis

5.1 Capital Expenditure (CAPEX) Breakdown

The total capital expenditure for the Sahara Neural Hub is estimated between \$611 million and \$1.135 billion for a fully built 50 MW facility including 2,000 NVIDIA H100 GPUs. The infrastructure component (buildings, power, cooling, networking) accounts for approximately \$281-575 million, while the IT equipment component (GPU servers, InfiniBand networking, storage) accounts for \$330-560 million. The following table provides a detailed breakdown of each CAPEX category, with low and high estimates reflecting different design choices and procurement strategies. It is important to note that GPU pricing is particularly volatile, and the emergence of newer architectures (B200, GB200) may shift the cost structure significantly. Additionally, the Dakhla Free Zone provides customs duty exemptions on all imported equipment, which can represent savings of 0-40% on equipment costs depending on the tariff classification, potentially saving tens of millions of dollars on the total project cost.

Component	Low (\$M)	High (\$M)	Notes
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Table 8: Capital Expenditure Breakdown (50 MW Facility)

5.2 Operating Expenditure (OPEX) Projection

Annual operating expenditure for the 50 MW facility is projected at \$82-99 million, with electricity representing the largest single cost item at approximately \$66 million per year. This calculation assumes a 50 MW IT load with a PUE of 1.3 (65 MW total input), 8,760 operating hours per year, and Morocco's business electricity rate of \$0.116/kWh. However, the use of self-generated renewable energy could significantly reduce this cost: at an LCOE of \$0.025-0.035/kWh for Dakhla wind energy, the electricity cost could drop to approximately \$14-20 million per year for the self-generated portion, representing annual savings of \$40-50 million compared to grid electricity. Staff costs in Morocco are 60-80% lower than in Western Europe, with data center technicians earning \$12,000-20,000/year and senior engineers \$20,000-40,000/year. A 40-person team would cost approximately \$1.2 million annually. Maintenance is budgeted at 3% of infrastructure CAPEX (\$8.4-17.3 million/year), insurance at 0.4% of total asset value (\$2.4-4.5 million/year), and bandwidth/connectivity at \$2-5 million/year for multiple 100G links.

Cost Item	Annual Cost (\$M)	Notes
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Table 9: Annual Operating Expenditure Projection

5.3 Revenue Model and ROI Analysis

The revenue model for the Sahara Neural Hub combines three primary streams: GPU-as-a-Service (GPUaaS), colocation services, and sovereign cloud services. GPUaaS is the highest-margin offering, with 2,000 H100 GPUs at full utilization generating \$35-120 million annually based on prevailing cloud rates. Colocation services at \$8-15/kW/month for a 50 MW facility would generate \$4.8-9.0 million/month (\$58-108 million/year) at full occupancy. Sovereign cloud services, including AI inference endpoints, model training platforms, and data lake services for African enterprises and governments, command premium pricing due to regulatory compliance requirements. The break-even analysis for GPU hardware is compelling: compared to AWS on-demand pricing (\$6.88/GPU-hour), owning an 8-GPU DGX H100 system at \$300K-\$450K breaks even in 9-14 months at 100% utilization. Even compared to neo-cloud pricing (\$2.50/GPU-hour for Spheron H100 SXM5), the payback period is 24-36 months. The project IRR is projected at 12-18% over a 15-year horizon, assuming 70% average utilization of GPU resources and 85% colocation occupancy, with the higher end achievable through aggressive renewable energy self-generation and premium sovereign cloud pricing. The Dakhla Free Zone tax regime (0% IS for 5 years, then 15% vs. standard 31%) provides a significant boost to after-tax returns.

5.4 Financing Strategy

The financing strategy leverages a combination of equity, green bonds, development finance, and project finance debt. The Dakhla Free Zone status facilitates foreign investment with full currency convertibility and unrestricted profit repatriation. Potential financing sources include: the Green Climate Fund (GCF), which has accredited MASEN as an implementing entity and targets 3,000 MW of additional renewable capacity; the World Bank (\$500M package for Morocco); the African Development Bank (SEFA program, ENR Africa facility); the European Investment Bank (EUR 300M financing to ONEE for grid strengthening in 2025); KfW (German Development Bank, historical partner for Moroccan ENR projects); and AFD (French Development Agency). Morocco has signed double taxation treaties with over 60 countries and bilateral investment treaties with over 70 countries, providing robust legal protection for foreign investors. The investment incentive framework includes up to 30% investment subsidy for strategic projects under Morocco's Investment Charter. Corporate loan interest rates in Morocco range from 5-7% for investment loans, with subsidized rates of 2-4% available through government programs. Green bond issuances by Moroccan banks (Attijariwafa Bank, BCP) provide an additional financing avenue at favorable terms.

6. Legal and Regulatory Framework

6.1 Dakhla West Africa Free Zone (DWAFZ)

The Dakhla West Africa Free Zone (DWAFZ), also known as Dakhla Atlantique Free Zone, operates under the legal framework established by Dahir 1-95-1 (Law 19-94 on export free zones), as amended by Law 57-19 modernizing the ZFE framework into Zones d'Acceleration Industrielle (ZAI). The zone encompasses 1,000 hectares of industrial land with an additional 2,650 hectares of industrial zone adjacent to the new Dakhla Atlantic Port (delivery 2028). Eligible sectors include renewable energy, tourism and services, research and innovation, logistics and maritime industries, specialized training, and export-oriented industries. The establishment procedure is streamlined through the CRI Dakhla one-stop shop, with typical implantation timelines of 2-6 weeks for complete dossiers. Since 2018, companies in the ZAI may sell up to 15% of their prior-year export turnover on the Moroccan domestic market, subject to standard customs duties and VAT. An important clarification: contrary to widespread belief, there is no general fiscal exemption regime for the Southern Provinces. The only substantive tax advantages are those provided by the ZAI/DWAFZ framework. This was confirmed by the Moroccan General Tax Administration and expert accountants in a June 2024 Challenge.ma investigation, which established that a 1970s-era royal statement about fiscal exemptions was never codified into law.

Tax	DWAFZ Advantage	Duration	Standard Rate
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Table 10: DWAFZ Tax Advantages vs. Standard Regime

6.2 Company Formation and Land Acquisition

Company formation in Morocco is streamlined through the CRI one-stop shop system, reformed by Law 47-18. The recommended legal form for this project is a Societe Anonyme (SA) or Societe par Actions Simplifiee (SAS), both offering limited liability and flexibility for large-scale investment. The SA requires a minimum capital of 300,000 MAD (approximately EUR 28,000) and at least 5 shareholders, while the SAS has no minimum capital requirement and can be formed with a single shareholder (SASU). The registration process involves 7 steps:

negative certificate (name verification), statutes drafting and signing, capital blocking in a bank, statutes registration with tax authorities, publication in the Official Bulletin, trade registry registration, and CNSS affiliation. The typical timeline is 2-6 weeks via the CRI. For land acquisition, Dakhla offers several pathways: private market purchase (recommended for titled land in PK40/Al Argoub), administrative concession from the Domaine National (through the Regional Directorate of National Domain), long-term lease (bail emphyteotique, 18-99 years), and allocation within the DWAFZ. Foreign investors can acquire non-agricultural land in Morocco without restriction, though properties in military or strategic zones require Interior Ministry authorization. It is critical to verify the existence of a registered land title (titre foncier) through the ANCFCC before any purchase, as informal land rights (melk, collective) do not provide the same legal security as a registered title.

6.3 Data Protection and Telecom Regulations

Operating a data center in Morocco requires compliance with two primary regulatory frameworks. Loi 09-08 (2009) on personal data protection, enforced by the CNDP (Commission Nationale de controle de la protection des Donnees a caractere Personnel), requires prior declaration before any personal data processing, mandates data security measures, and governs cross-border data transfers (allowed if the destination country provides equivalent protection or with CNDP authorization). Morocco does not currently hold an EU adequacy decision, meaning that data transfers from the EU require Standard Contractual Clauses (SCC) or Binding Corporate Rules (BCR) from the European data exporter's side. The telecom regulatory framework, governed by Loi 24-96 and Loi 55-01, is administered by the ANRT. A data center providing only hosting/colocation services (without public telecom services) generally requires only a prior declaration to the ANRT. However, if the facility provides connectivity or transit IP services, an ANRT authorization is required. The cybersecurity framework (Loi 05-20, 2020) establishes the CNDP (Cybersecurity), which sets national cybersecurity strategy and standards for critical infrastructure operators, including large data centers. Alignment with ISO 27001 and NIST frameworks is recommended and increasingly expected by institutional clients.

7. Risk Assessment

7.1 SWOT Analysis

Category	Factor	Impact
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Table 11: SWOT Analysis - Sahara Neural Hub

7.2 Key Risk Factors and Mitigation

Geopolitical Risk: Dakhla is located in the Western Sahara, a territory subject to an ongoing sovereignty dispute between Morocco and the Polisario Front. While Morocco exercises de facto administrative control and has accelerated investment under the New Development Model for the Southern Provinces, the geopolitical ambiguity may concern some international investors. Mitigation: The Moroccan government has demonstrated strong commitment to infrastructure investment in the region (\$1.2-1.6B new port, 2.4B MAD grid connection, 2.5B

MAD desalination), providing de facto assurance of continued support. The AU and numerous countries have recognized Moroccan sovereignty, and the UN political process continues to evolve favorably.

Infrastructure Risk: The Dakhla-Casablanca 400 kV transmission line (3 GW capacity) is critical for reliable power supply and renewable energy evacuation. Delays in this project could limit the facility's ability to operate at full capacity or export surplus renewable energy. Mitigation: The project is a national priority with 5 shortlisted candidates as of February 2025 and expected delivery by 2028. In the interim, on-site BESS and diesel backup generators can ensure power continuity, and the 60 MW wind farm already operational provides a baseline.

Market Risk: The facility depends heavily on export demand (GPUaaS to European and pan-African clients) as the local Dakhla market is negligible. If hyperscale cloud providers expand their own African data center capacity (AWS, Azure, Google), the competitive landscape could shift. Mitigation: The sovereign cloud and data localization angle provides differentiation that hyperscalers cannot easily replicate. The first-mover advantage in Dakhla creates barriers to entry, and the green energy narrative resonates with ESG-conscious institutional clients.

Technology Risk: GPU technology evolves rapidly, with NVIDIA releasing new architectures every 18-24 months. The H100/B200 generation may be superseded by the time Phase 2 is operational, potentially requiring additional CAPEX for hardware refresh. Mitigation: The modular data center design allows for technology refresh within existing rack infrastructure. GPUaaS pricing historically tracks hardware generations, so newer hardware commands premium rates. The facility's flexible cooling and power design (up to 140 kW/rack) accommodates next-generation hardware.

8. Major Infrastructure Projects in Dakhla

The Moroccan government has committed to an unprecedented infrastructure investment program in the Dakhla-Oued Ed-Dahab region, signaling strong commitment to the economic development of the southern provinces. These mega-projects create a favorable ecosystem for a data center investment by addressing key constraints around power, water, connectivity, and logistics. The following table summarizes the major infrastructure projects currently underway or recently completed, which collectively represent an investment exceeding \$3 billion and will fundamentally transform Dakhla's infrastructure capacity over the 2025-2030 period.

Project	Investment	Delivery	Status
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Table 12: Major Infrastructure Projects in Dakhla-Oued Ed-Dahab

9. Competitive Landscape

The competitive landscape for data center investment in Morocco and Africa is rapidly evolving. Within Morocco, three major projects have been announced in 2024-2025: the Iozera facility in Tetouan (386 MW, \$500M MoU, targeting the European AI market with proximity to Gibraltar), the Naver Cloud facility also in Tetouan (500 MW, partnership with NVIDIA, Korean cloud giant entering Africa), and the Nouaceur project near Casablanca (500 MW target, 100% renewable, led by TAQA Morocco). These projects collectively represent 1,386 MW of announced capacity, though none are yet operational as of early 2026. The key differentiator for the Sahara Neural Hub in Dakhla is its unique positioning as a sovereign African data storage facility with direct submarine cable connectivity to West Africa, which the Tetouan projects (oriented toward European clients) cannot replicate. In the broader African context, South Africa's Teraco leads with ~130 MW operational and expansion underway, while Nigeria's MDXi (Equinix) and Kenya's emerging IXAfrica (53 MW announced) represent the primary competitive hubs. However, none of these competitors combine the green energy narrative, West African connectivity, and sovereign cloud positioning that Dakhla offers.

Project / Operator	Location	Capacity	Status / Differentiator
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Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Teraco' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Teraco', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Johannesburg, SA' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Johannesburg, SA', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '~130 MW' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='~130 MW', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Largest in Africa; expanding' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Largest in Africa; expanding', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'MDXi (Equinix)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='MDXi (Equinix)', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Lagos, Nigeria' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Lagos, Nigeria', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Tier III' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Tier III', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Leading West Africa hub' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Leading West Africa hub', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'IXAfrica' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='IXAfrica', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Nairobi, Kenya' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Nairobi, Kenya', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '53 MW' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='53 MW', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '2nd hyperscale announced Nov 2024' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='2nd hyperscale announced Nov 2024', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

<pre>Paragraph('caseSensitive': 1 1 'encoding': 'utf8' 'text': 'Sahara Neural Hub' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Sahara Neural Hub', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Dakhla, Morocco' 'frags': [ParaFrag(__tag__='para', bold=0, font Name='LiberationSerif' , fontSize=9, greek=0, italic=1, link=[], rise=0, text='Dakhla, Morocco', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '50-100 MW' 'frags': [ParaFrag(__tag__=' para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='50-100 MW', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Green + Sovereign + West Africa gateway' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Green + Sovereign + West Africa gateway', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>
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Table 13: Competitive Data Center Projects in Morocco and Africa

10. Implementation Roadmap

The implementation of the Sahara Neural Hub follows a phased approach designed to manage risk, secure early revenue, and align with the delivery of critical infrastructure projects (particularly the Dakhla-Casablanca THT line in 2028). The roadmap is structured in three phases spanning 36-60 months from project initiation to full build-out, with each phase designed to be independently viable and revenue-generating. This phased approach allows for course corrections based on market response, technology evolution, and infrastructure development timelines.

Phase	Timeline	Scope	Key Milestones
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Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Phase 0: Foundation' 'frags': [ParaFrag(__tag__='p ara', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Phase 0: Foundation', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Months 1-6' 'frags': [ParaFrag(__tag__='p ara', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Months 1-6', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Legal, permits, land' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Legal, permits, land', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'SAS formation, DWAFZ registration, land acquisition, environmental study' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='SAS formation, DWAFZ registration, land acquisition, environmental study', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Phase 1A: Energy' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Phase 1A: Energy', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Months 6-24' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Months 6-24', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Renewable energy + grid' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Renewable energy + grid', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Wind farm 50 MW, solar 30 MW, BESS 100 MWh, grid connection agreement' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Wind farm 50 MW, solar 30 MW, BESS 100 MWh, grid connection agreement', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Phase 1B: DC Shell' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Phase 1B: DC Shell', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Months 12-30' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Months 12-30', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Building + infrastructure' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Building + infrastructure', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Shell construction, power infrastructure, cooling, UPS, networking' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Shell construction, power infrastructure, cooling, UPS, networking', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Phase 1C: IT Deploy' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Phase 1C: IT Deploy', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Months 24-36' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Months 24-36', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'GPU clusters Phase 1' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='GPU clusters Phase 1', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '500 H100/B200 GPUs, InfiniBand fabric, sovereign cloud platform launch' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='500 H100/B200 GPUs, InfiniBand fabric, sovereign cloud platform launch', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Phase 2: Scale' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Phase 2: Scale', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Months 36-60' 'frags': [P araFrag(__tag__ ='para', bold=0, f ontName='Liber ationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Months 36-60', textColor =Color(.117647, .121569,.129412 ,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Full build-out' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Full build-out', textColor=Color(.1 17647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Additional 1,500 GPUs, expand to 50 MW, seawater cooling eval, 100 MW planning' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Additional 1,500 GPUs, expand to 50 MW, seawater cooling eval, 100 MW planning', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>
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Table 14: Implementation Roadmap

11. Key Contacts and Institutions

The following table provides the essential contacts for advancing the Sahara Neural Hub project, including government agencies, investment promotion bodies, regulatory authorities, and key private sector stakeholders. These contacts have been verified through official websites and public directories as of March 2026. It is recommended to initiate engagement with the CRI Dakhla as the first point of contact, as the center provides a one-stop shop for investor facilitation including company registration, land allocation, and regulatory guidance.

Organization	Role	Contact
<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CRI Dakhla' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='Libera tionSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='CRI Dakhla', textColor=Color(.117647,. 121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Investment facilitation (one-stop shop)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Investment facilitation (one-stop shop)', textColor=Color(.117647,. 121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '+212 5 28 89 85 35 / contact@cri-dakhla.ma' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='+212 5 28 89 85 35 / contact@cri-dakhla.ma', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'AMDI / AMDIE' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='AMDI / AMDIE', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'National investment promotion' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='National investment promotion', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '+212 5 22 26 88 88 / contact@amdie.gov.ma' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='+212 5 22 26 88 88 / contact@amdie.gov.ma', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'ANRT' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='ANRT', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Telecom regulation / DC licensing' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Telecom regulation / DC licensing', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '+212 5 37 71 84 00 / anrt@anrt.ma' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='+212 5 37 71 84 00 / anrt@anrt.ma', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CNDP (Data Protection)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='CNDP (Data Protection)', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Personal data regulation (Loi 09-08)' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Personal data regulation (Loi 09-08)', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '+212 5 37 72 44 00 / contact@cnpd.ma' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='+212 5 37 72 44 00 / contact@cnpd.ma', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Wilaya de Dakhla' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Wilaya de Dakhla', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Regional administration' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Regional administration', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '+212 5 28 89 60 00' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='+212 5 28 89 60 00', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'CCIS Dakhla' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='CCIS Dakhla', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Chamber of Commerce' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='Chamber of Commerce', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '+212 5 28 89 65 65' 'frags': [ParaFrag(__tag__='para', bold=0, fontName='LiberationSerif', fontSize=9, greek=0, italic=1, link=[], rise=0, text='+212 5 28 89 65 65', textColor=Color(.117647,.121569,.129412,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
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Table 15: Key Contacts and Institutions

12. Conclusion and Recommendation

The Sahara Neural Hub project represents a compelling investment opportunity at the intersection of three mega-trends: the explosive growth of AI computing demand, the acceleration of data sovereignty requirements across Africa, and the global transition to green digital infrastructure. Dakhla's unique combination of world-class wind resources (LCOE \$0.022-0.035/kWh), operational submarine cable connectivity to Europe and West Africa (20-40 Tbps), favorable tax regime (DWAFZ: 0% IS for 5 years, then 15%), and exceptional free cooling potential (5,000-7,500 hours/year) creates a value proposition that no other African location can currently replicate in its

entirety.

The financial analysis indicates a total CAPEX of \$611M-\$1.135B for a 50 MW facility with 2,000 GPU servers, with annual OPEX potentially as low as \$30-53M when leveraging self-generated renewable energy versus \$82-99M on grid power. The project IRR of 12-18% over a 15-year horizon, combined with the Dakhla Free Zone tax advantages, presents an attractive risk-adjusted return profile. The phased implementation strategy (6-month foundation, 30-month Phase 1, 60-month full build) manages execution risk while aligning with the delivery of critical infrastructure projects, particularly the Dakhla-Casablanca 3 GW transmission line expected in 2028.

The key risks (geopolitical ambiguity, infrastructure dependency, and technology evolution) are manageable with appropriate mitigation strategies. The first-mover advantage in Dakhla is time-sensitive: as Morocco's data center pipeline matures and other regions develop sovereign cloud capacity, the window of opportunity for establishing Dakhla as the definitive gateway between Europe and West Africa will narrow. The recommendation is to proceed with Phase 0 (Foundation) immediately, targeting SAS formation, DWAFZ registration, and land acquisition within 6 months, with a Phase 1 investment decision contingent on confirmation of the THT line construction timeline and the securing of anchor tenant commitments for at least 30% of Phase 1 capacity. This project has the potential to position Morocco as the continental leader in sovereign AI infrastructure and to establish Dakhla as a landmark green data center destination that sets the standard for sustainable digital infrastructure in the 21st century.

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